

UMME KULSOOM A

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SKILLS

Programming Languages	Java, Python, C, HTML
Data Analytics	Power BI, Tableau, Pandas
Databases	MySQL
AI/ML	Machine Learning, Computer Vision, NLP, Generative AI
Core Subjects	OOPS, DBMS, Computer Networks, JDBC, JSP

PROJECTS

Reader's Lane – E-Commerce Bookstore Web Application Apr 2026 - May 2026
Java, JSP, Servlets, JDBC, MySQL, HTML, CSS, JavaScript, Apache Tomcat

- Developed a full stack e-commerce bookstore application using Java, JSP, Servlets, JDBC, MySQL, HTML, CSS, and JavaScript following MVC architecture.
- Implemented authentication, session management, cart handling, and CRUD-based admin functionalities with deployment on Apache Tomcat Server.

ML-Driven Livestock Welfare System May 2025 - Dec 2025
Python, PyTorch, DeepLabCut, LSTM, MobileNet, OpenCV Livestock System

- Created AI system for early detection of lameness and mastitis in cattle, integrating computer vision and deep learning.
- Implemented DeepLabCut+LSTM for Lameness; RGB-image classification using Mobile-net for Mastitis.
- Building a low-cost, scalable deployment pipeline aimed at real-time monitoring in rural farms

Real-Time Soil Moisture Monitoring System Sept 2024 - Feb 2025
C++, JS, ESP32, Soil Moisture Sensors, MQTT, WebSockets, IoT, Weather APIs Soil Moisture

- Built an IoT-based system using ESP32, soil sensors, and MQTT for real-time soil monitoring and automated water control.
- Integrated WebSocket communication for sub-2s latency and developed predictive dashboards using weather APIs.

WORK EXPERIENCE

Java Full Stack Development Intern - Tap Academy Jan 2026 - Present

- Working on real-world full stack web development concepts including frontend design, backend processing, database integration, and MVC-based application architecture.
- Applying Java, JSP, Servlets, JDBC, MySQL, and Apache Tomcat to build dynamic web applications with secure authentication and efficient request handling.

Data Processing & Machine Learning Intern - Elevate Labs Apr 2025 - May 2025

- Performed data preprocessing, EDA, and implemented ML models including regression, SVM, clustering, and logistic regression.
- Developed an AI-powered NLP story generator pipeline using Streamlit for automated content generation.

IoT and Web Development Trainee - Samsung Innovation Campus Sep 2024 - Feb 2025

- Collaborated under industry mentorship to develop an IoT-based irrigation and soil monitoring system.
- Worked on system integration, testing, sensor communication, and performance optimization for automated irrigation workflows.

ACHIEVEMENTS

- Secured **2nd Prize** at the **IEEE Project Presentation Competition (KAGADA 2025)** held at **UVCE**.
- Selected among **India's Top 10 Teams** in **Hack to Hire** and presented the solution at national level.
- Authored and published a **conference research paper** in **IEEE**.

EDUCATION

2022 - 2026 **B.E. in Artificial Intelligence & Machine Learning** (CGPA: 9.2)
Cambridge Institute of Technology, Bangalore

2020 - 2022 **Pre-University Course (PCMC)** (95%)
St. Charles Women's PU College, Bangalore